

UNIVERSITY OF WOLLONGONG IN DUBAI

Accessibility Scene Hazard Assistant

An Integrated Computer Vision Application for Accessible Scene Guidance

**CSCI435 - Computer Vision Algorithms and Systems
Spring 2026**

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Report date: 27 June 2026

Report at a glance

We created and developed a local deployable web application that combines the following: image enhancement, edge detection, object proposal, custom-trained object recognition, moving-object detection, and object tracking all in one workflow. The system should be able to receive an uploaded image or a webcam snapshot or a video and return the results in the form of an annotated image, text instructions, latency and estimated throughput.

The notebook's performance was 93% avg precision, 94.6% recall, and 93.1% F1 under a mean of 7 different scene conditions. The 50.1 FPS (20ms) full still-frame pipeline averaged was achieved, meeting the 10 FPS requirement.

Abstract

This project explores the possibility of combining multiple computer vision techniques together as an integrated and meaningful system and not as individual and disconnected lab exercises. Our idea is a simple scene assistant, which will be able to recognize three types of markers: stop, warning and safe route. A customizable dataset of 560 marker crops is used and created in the notebook. The performance of RBF support vector machine, random forest and k-nearest neighbor (kNN) classifiers are compared using HOG, HSV, Intensity and Hu moment. Then a selected SVM is combined with color proposals, adaptive Canny edges, CLAHE enhancement, MOG2 background modelling and centroid tracking. Held-out recognition, class-aware scene detection, lighting and orientation changes, noise, blur, occlusion, clutter, blank inputs and CPU speed. Results were very good in this controlled synthetic environment, but there are weaknesses to be addressed between that and real life, and the need for user-testing prior to any real deployment.

Submission components

- Executed Jupyter notebook: CSCI435_Project.ipynb.
- Dependency file and reproducible execution instructions in README.md.
- Two-minute video demonstration from actual pipeline outputs.
- Editable DOCX report and submission-ready PDF report.
- Timed defence guide, robust sample inputs and requirement-to-evidence matrix.

Introduction

Problem statement

Those whose vision is impaired or sub-optimal may require an alert to any warning, a reminder of a route cue, identification of some visible boundaries and a visual awareness of movement in the camera view. Many of the demo images that are typically processed in class are just processed with one algorithm, and that fails to demonstrate how an actual working application should check the validity of the input, combine the results, or communicate uncertainty. With this in mind, we aimed to establish a unified interface between low-level, mid-level and high-level vision tasks based on a unified accessibility user story.

User story

User story. As end user, I want to upload an image or a video or capture an image with my webcam, and then there is an interface that is better at enhancing an image (scene) in the frame on screen; there is an interface that is better at identifying objects that could be markers in the image (scene) on screen; there is an interface that is better at showing boundaries within the image (scene) on screen; there is an interface that is better at showing movement within the image (scene) on screen; and there is an interface that provides concise instructions.

Selected vision capabilities:

- The image was enhanced using gray-world colour balancing and CLAHE in the luminance channel.
- Edge detection: Adaptive Canny thresholds, morphological closing and a visible edge overlay.
- Object detection: HSV colour segmentation, morphology, contour filtering and bounding boxes.
- The RBF SVM object recognition module: RBF SVM that was learned from the group's generated custom marker database.
- Change and moving object detection: MOG2 adaptive background modelling on video frames.
- Centroid association with persistent tracking numbers, missing frame tolerance.

Scope and safety

The app is not a registered safety device, but rather an academic proof-of-concept. It uses a restricted vocabulary and a synthetic evaluation, which are clearly and intentionally limited for the purpose of allowing all steps in training and testing to be reproduced. To avoid use and legal problems, representative data, and a much more conservative false-negative analysis, requiring calibrated depth, accessibility studies and privacy review, are required for real-world use..

System architecture

Accessibility Scene Hazard Assistant - System Architecture

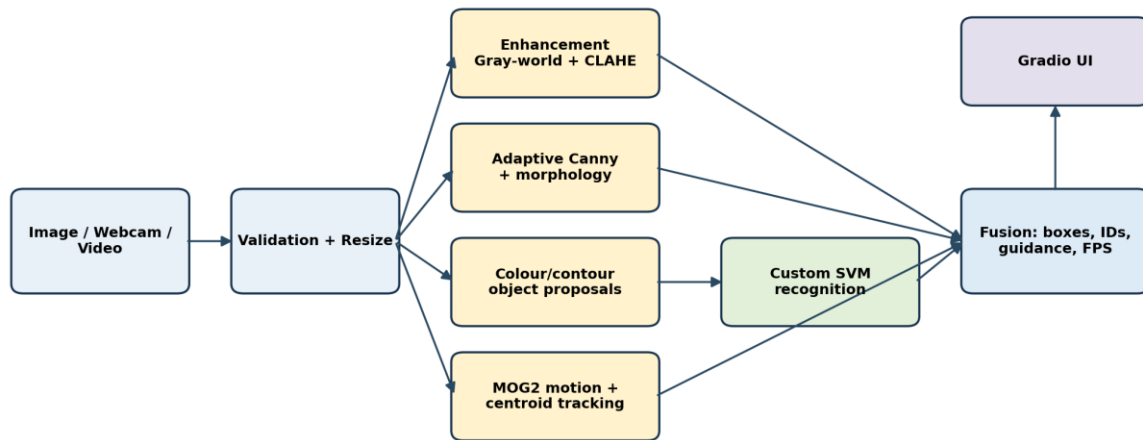


Figure 1. System architecture and data flow.

The front-end and back-end are both included in the notebook. The browser interface is provided by Gradio, and the layers of processing and models are provided by Python, OpenCV and scikit-learn. An input is checked and shrunk to maximum width of 640 pixels. Enhanced frames offers color-proposal and edge branches. The custom SVM is used to classify the candidate pixels. The centroid tracker and MOG2 process the video frames. Fusion takes boxes, labels and track IDs and constructs a text guidance sentence and reports the resulting latency/FPS.

Input modalities and frontend:

You can upload an image or take a picture with the webcam and the Image / Webcam tab will display 4 outputs: an annotated image (with the integrated tool), an enhanced image, an edge view and a guidance view. Video tab would also support an uploaded file, take the same frame pipeline, add motion/tracking, and provide an annotated video and overall statistics. Both paths respect the need for at least two

modalities of input and maintain consistency of the interface.

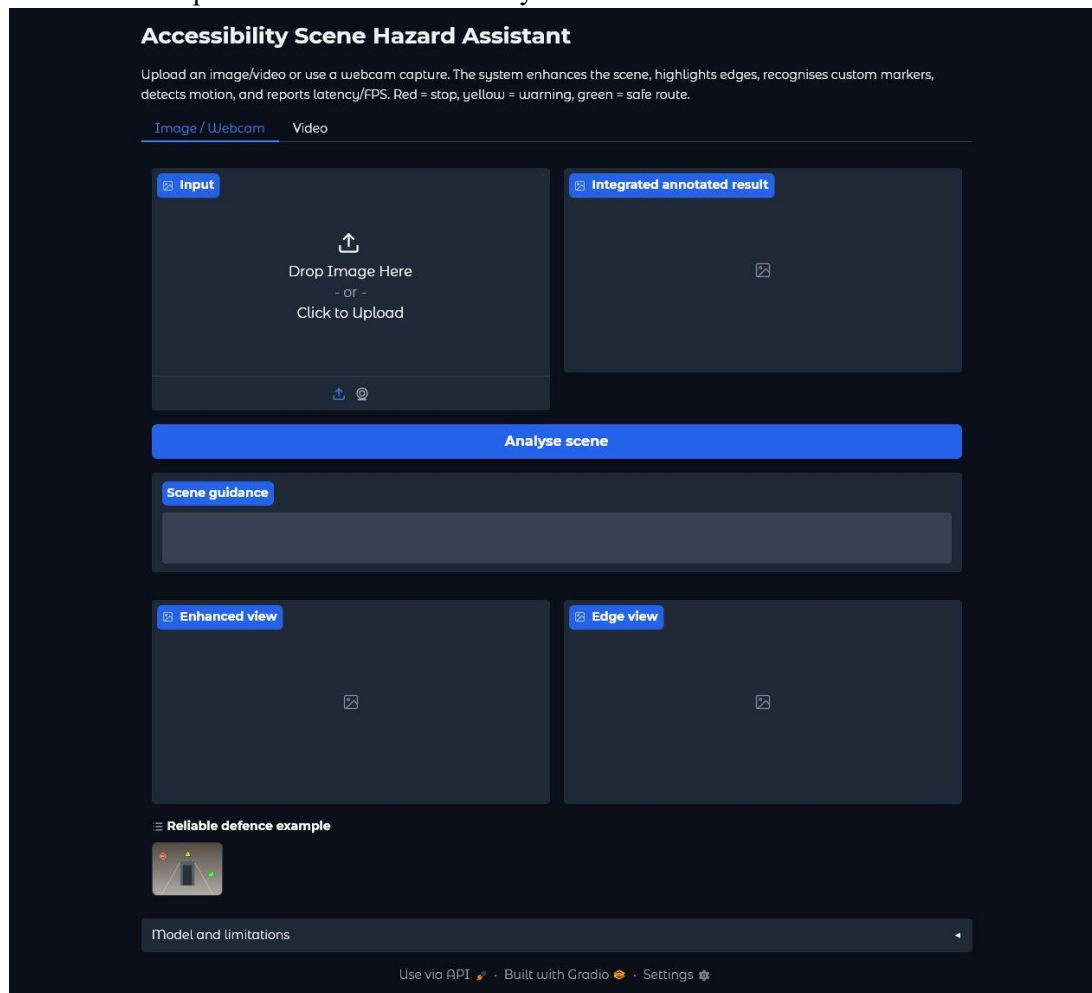


Figure 2. Verified local Gradio interface in the Image / Webcam mode.

Implementation details

Enhancement and edge processing:

Gray-world balancing calculates a target mean for all the BGR channels and reduces channel scaling to prevent large colour changes. The image is then transformed into CIELAB where only the luminance component is subject to CLAHE. A 3 x 3 closing operation is then applied to join short breaks and the adaptive Canny thresholds are computed from the median of a grayscale image after Gaussian smoothing. This branch provides a clear boundary view and does not explicitly take over the real meaning. A formal proposal for the creation of an object.

Object proposal and custom recognition:

Red, yellow and green proposal masks are created using HSV thresholds. Opening deletes isolated pixels, closing fills small gaps. Original and enhanced pixels are both looked for with separate colour masks, so that no dominant single colour will be lost over balancing and adjacent marker colours don't mix.

Contours with less than 300 pixels or those that fall outside of the wide aspect-ratio range are discarded. Every other object is placed at a fixed foreground size and centered in a neutral canvas of size 96 x 96. This canonicalisation helped address an important training/inference mismatch, namely that the recogniser is seeing the same scale of the object in both training and inference. The SVM collects the proposal masks in accordance with the appearance distribution in the training phase using pre-enhanced validated pixels. If the synthetic crop model is in doubt, a conservative fusion rule verifies real evidence: red octagonal shapes signal stop, yellow triangular shapes signal warning, and green circular shapes with bright symbol in the center indicate safe. This recovered examples without updating the learned model.

```
enhanced = enhance_image(frame)
edges = adaptive_canny(enhanced)
detections = detect_markers(
    enhanced, self.model, classification_frame=frame
)
```

Motion detection and tracking

For video, OpenCV's MOG2 model will learn the adaptive background. A high binary threshold is used to remove shadow valued pixels, followed by opening and dilation. The contours between 500 to 65% of the area of the frame are moving in 4 warm up frames. A light centroid tracker connects the current box to the closest existing centre, allocates new IDs if needed and deletes IDs after 8 unobserved frames. This approach is simple and quick to explain, but has the disadvantage that it can switch identities when objects overlap.

Input validation and robustness controls

Fails to accept empty, bad, and invalid input images and gives readable error messages.

- Converts grayscale and four channel images to three channel BGR.
- Reduces wide images to maintain aspect ratio, reduces latency, reduces memory usage.
- Displays a message without a marker for images that are empty or do not match.
- Exports video file as MP4 (falls back to AVI if local codec isn't initialized)

Custom data and model training

Dataset construction

The notebook produces 560 labelled examples: 140 examples in each of the classes: stop, warning, safe and other. Variations are background level gradient, scale, rotation, brightness, blur, Gaussian noise and partial occlusion. The other class is a red square with a diagonal line, which was deliberately made to

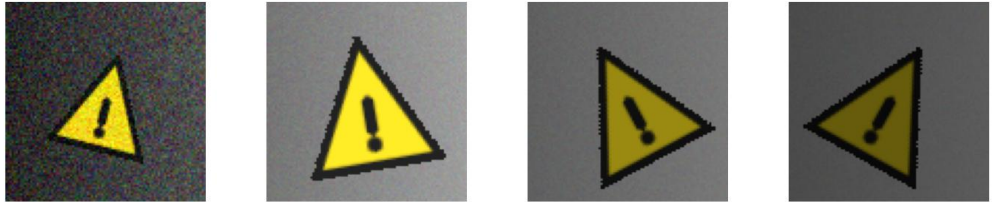
successfully pass the colour proposal phase and be recognized using shape evidence. Random seeds are

Custom synthetic dataset samples

Stop



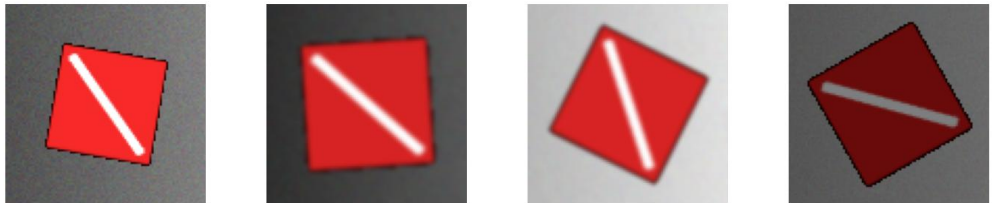
Warning



Safe



Other



set for repeatability.

Figure 3. Examples from the custom synthetic dataset.

Features and candidate models

For each of the canonical crops, HOG gradients are included in addition to 18-bin hue and 12-bin saturation histograms, 4 intensity/colour statistics and 7 Hu moments on a log scale. A 75/25 split was performed and three lightweight classifiers were compared against the same samples. Certainly, the scaling is performed within the SVM and k-NN pipelines to prevent leakage of information from the held out split.

Model	Validation accuracy	Training (s)	Inference (ms/crop)
SVM (RBF)	100.0%	0.105	0.070
Random forest	100.0%	0.168	0.193
k-NN	100.0%	0.005	9.172

Table 1. Candidate-model comparison from the executed notebook.

Model choice

Since all three models were able to correctly classify all the controlled crop data, crop accuracy was not a sufficient reason to make a selection. We chose the RBF SVM due to the non-linear boundary, class balancing, low per-crop inference time and probability output for user confidentiality. The trained pipeline is saved as a joblib artifact and is also reconstructed from the source when the notebook is executed.

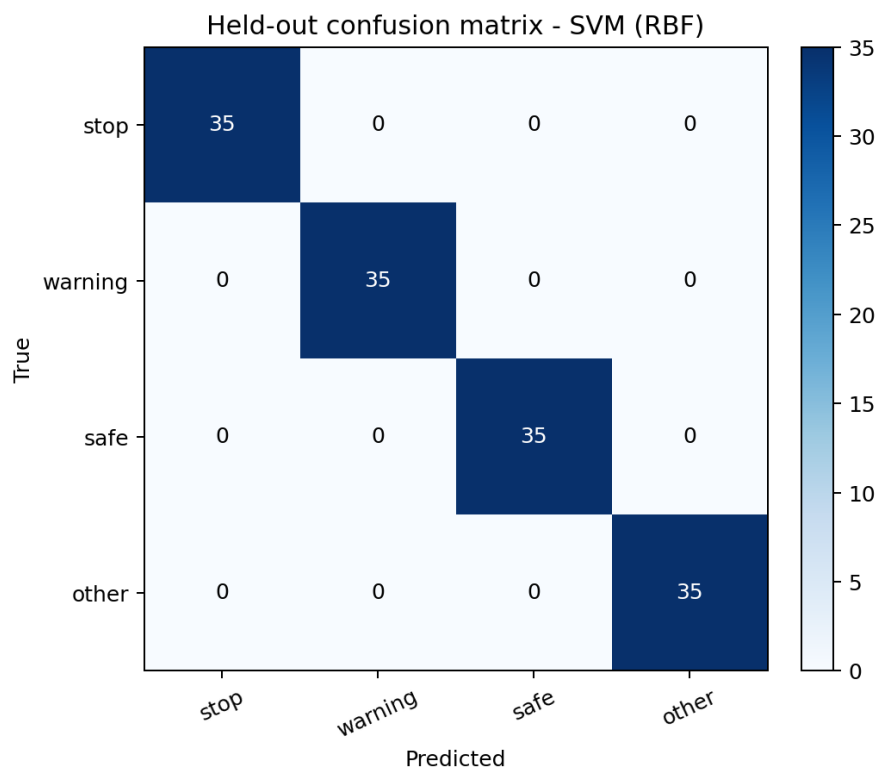


Figure 4. Held-out confusion matrix for the selected SVM.

Interpretation warning. The 100% held-out crop accuracy applies only to the generated marker distribution. It must not be presented as real-world safety accuracy. End-to-end scene metrics are more demanding and are reported separately.

Experiments and results

Evaluation protocol

The number of crops per class and condition for recognition robustness was 35. For end to end detection 12 scenes per condition were used, one to three target classes each scene, and a random sample of seeds not used for training. Prediction was considered true positive if class matched and intersection-over-union to an unused ground-truth box was ≥ 0.30 . Precision, recall, F1 and mean matched IoU were subsequently combined. Seven scene conditions consisted of bright/dim, rotated, blurred, noisy, occluded and cluttered.

Condition	Precision	Recall	F1	Mean IoU	Recognition accuracy
Bright	100.0%	100.0%	100.0%	0.765	100.0%
Dim	95.5%	87.5%	91.3%	0.727	100.0%
Rotated	100.0%	100.0%	100.0%	0.770	94.3%
Blurred	100.0%	100.0%	100.0%	0.803	100.0%
Noisy	100.0%	100.0%	100.0%	0.752	100.0%
Occluded	95.0%	79.2%	86.4%	0.724	100.0%
Cluttered	60.5%	95.8%	74.2%	0.750	N/A

Table 2. Condition-wise robustness results from the executed notebook.

The mean precision, recall, and f1 values across the seven conditions were 93.0%, 94.6%, and 93.1%, respectively. In this run, generated scenes were flawlessly detected both in terms of clarity and noise. Rotated, occluded and cluttered scenes were more difficult. Saturated distractors reduced precision, while rotated and occluded targets mainly reduced recall, due to clutter. This pattern is consistent to the proposal/classification design.

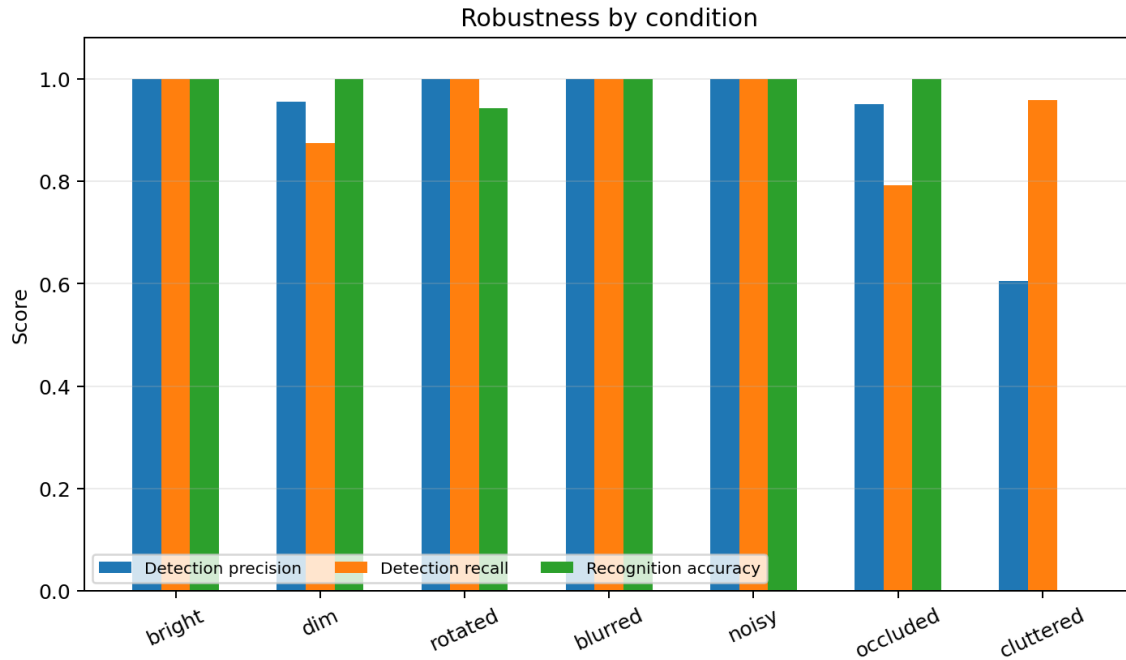


Figure 5. Detection and recognition robustness by condition.

Performance

Measure	Result
Still-frame mean latency	19.96 ms
Still-frame median latency	19.70 ms
Still-frame 95th-percentile latency	21.74 ms
Estimated still-frame throughput	50.09 FPS

Measure	Result
Processed-video mean latency	16.75 ms/frame
Processed-video estimated throughput	59.72 FPS
Rubric threshold	10 FPS - met

Table 3. CPU timing measured over the complete integrated pipeline.

The benchmark features the following functions: validation, resizing, improvement, edge, proposal generation, feature extraction, SVM inference, drawing and guidance. MOG2 and tracking are also included in the video timing. FPS is not equal to the refresh rate of the camera display, but it is equal to the reciprocal of processing latency, which indicates the processing capacity. The results vary depending on hardware, but the test run was significantly higher than the minimum criterion of 10 fps.

Qualitative results

Stop marker on the left (98% confidence); Safe-route marker on the right (96% confidence); Warning marker on the centre (92% confidence).

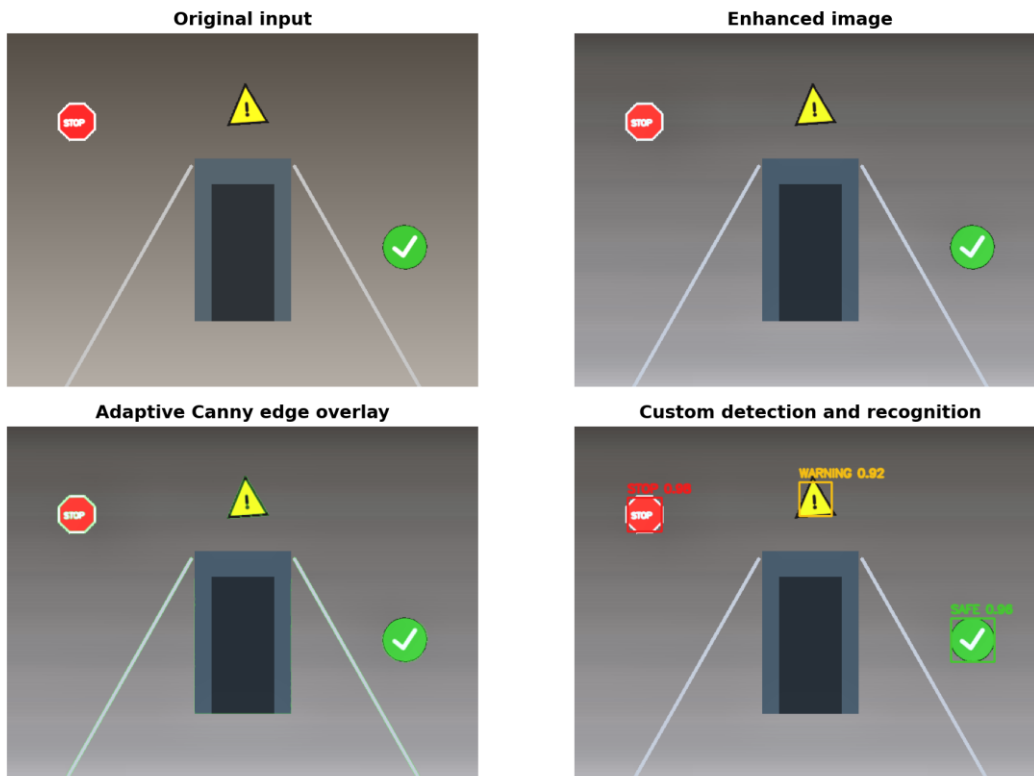


Figure 6. End-to-end image workflow and generated guidance.

The tasks complement each other as shown in the clean defence scene. The enhancement normalizes contrast, the edge view shows structural boundaries, colour/contour proposals localize the candidates, the SVM labels all three markers, and the fusion layer outputs: 'Safe-route marker on the right; warning marker in the middle; stop marker on the left. Video examples add the motion boxes and persistent ids with the same base pipeline.

Correctness and edge-case tests

- Blank scene of all markers: no marker detected and markers are not readable message.
- Small 12 x 12 picture: value exception thrown and picture not processed by the OpenCV function.
- Dark, rotated, blurry, noisy and partially occluded scenes: processed and scored based on their conditions.
- Cluttered scene: False positives are not excluded from the results.
- User provided real style regression scene: it was found that the scene where the user provided a stop sign was detected at 98.0% and the scene where the user provided a safe sign was detected at 94.6%.
- Processed the video successfully (90 frame), producing motion/outputs and tracking outputs.

Challenges and critical analysis

Training-to-inference mismatch

An early version classified padded scene crops directly even though training used tightly framed 96 x 96 examples. Crop accuracy looked perfect, but scene recall was only 7.1%. This was a useful warning against reporting a convenient metric without testing the complete system. Canonical foreground scaling and recognition from pre-enhancement pixels raised the verified mean scene F1 to the value reported in Table 2.

Remaining limitations

- Synthetic-to-real domain gap: natural signs, reflections, weathering and camera compression are not represented.
- Colour dependence: extreme colour casts or desaturation may prevent proposal generation before the SVM runs.
- No calibrated distance: left/centre/right guidance is relative image position, not metric depth or collision risk.
- Motion is not semantic: MOG2 identifies changing regions but cannot state whether they are people or vehicles.
- Simple tracking: nearest-centroid matching may exchange IDs during crossing or prolonged occlusion.
- Safety and accessibility: no target-user study, audio/haptic interface or formal false-negative safety threshold was completed.

UI and deployment trade-offs

Gradio made the application fast to reproduce and suitable for a live local defence. It also exposes upload and webcam sources without maintaining a separate JavaScript codebase. The trade-off is less control

over mobile layout and browser codec behaviour than a custom frontend. Local hosting avoids sending camera data to a cloud service, but the prototype does not yet implement persistent privacy controls or authentication.

Individual contributions

The following table records the agreed primary workstreams. Integration, review and defence preparation were shared so that each member contributes to coding, documentation and presentation as required.

Member	Student ID	Primary coding	Documentation	Defence role
Mehdi Leghmizi	8528834	UI integration and input handling	User story and requirements	Opening and UI demo
Neeraj Santosh	8329345	Custom dataset and model training	Training methodology	Model selection
Muhammad Soban	8555588	Enhancement, edges and robustness tests	Experiment methodology	Image robustness
Zachary Bracke	8947405	Motion, tracking and performance	Results and critical analysis	Video and performance
Mostafa Shalash	7391493	System integration and QA	Architecture, conclusion and references	Architecture and closing
Luis Vargas	7655538	Feature engineering & viz	Reproduction guide & appendix	Features & evaluation

Table 4. Group contribution record and defence responsibilities.

Conclusion and future work

The project demonstrates a working integrated system rather than a collection of isolated scripts. Six permissible computer vision capabilities operate inside one image/video workflow and are exposed through a responsive local web interface. The custom SVM satisfies the training requirement, the end-to-end evaluation tests robustness rather than relying only on crop accuracy, and the measured performance is comfortably above the real-time threshold. The most important lesson was that complete-system evaluation can reveal failures hidden by excellent component metrics.

Future work should collect consented real images and videos, annotate objects with bounding boxes, compare a fine-tuned detector with the current proposal-plus-classifier approach, calibrate depth, quantify false negatives by hazard severity and test audio/haptic guidance with target users. A mobile deployment would also need model compression, battery profiling, offline inference and privacy-by-design review.

References

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Appendix A - Rubric traceability

Criterion	Marks	Evidence
Four or more tasks	15	Six integrated capabilities in VisionSystem; Figures 1 and 6.
Correctness and robustness	15	Seven conditions, edge-case assertions and class-aware IoU evaluation; Table 2.
UI and UX	10	Verified Gradio image/webcam and video tabs; Figure 2.
Model selection and training	10	560 custom images, three-model comparison and saved SVM; Table 1.
Architecture and code quality	10	Typed functions/classes, unified notebook, README and architecture diagram.
Performance	5	50.1 FPS measured versus 10 FPS requirement; Table 3.
Report quality	15	Required sections, diagrams, quantitative results, qualitative examples and critical analysis.
Live demonstration	15	Local interface, stable generated examples, 2:05 backup video and defence guide.
Teamwork	5	Named coding, documentation and defence responsibilities; Table 4.

Appendix B - Reproduction and defence checklist

- Create a Python 3.12 virtual environment and install requirements.txt.
- Open CSCI435_Project.ipynb and run all cells from top to bottom.
- Confirm that the model comparison, confusion matrix, robustness table and performance assertion appear.
- Open the printed local Gradio URL and process the generated defence image and sample video.
- Show image enhancement, edge overlay, marker boxes, guidance, motion IDs and measured FPS.
- Keep artifacts/CSCI435_Demonstration_Video.mp4 available as the required video and technical-failure backup.
- Push the repository to GitHub or GitLab, grant the lecturer access and place the repository link in the Moodle submission.